EXPERIMENT 1: 8 QUEENS PROBLEM

AIM

To place 8 queens on an 8 × 8 chessboard so that no two queens attack each other; that is, no two queens share the same row, column, or diagonal.

PROCEDURE

1. Create an 8 × 8 board initialized with empty cells.

2. Place one queen in each row by checking whether the position is safe.

3. If a position causes a conflict, try another column; if none is safe, backtrack.

4. Stop and display the board when all eight queens have been placed safely.

CODE (PYTHON)

```
N = 8

def is_safe(board, row, col):
    for r in range(row):
        if board[r][col]:
            return False
        if col - (row - r) >= 0 and board[r][col - (row - r)]:
            return False
        if col + (row - r) < N and board[r][col + (row - r)]:
            return False
    return True

def solve(board, row):
    if row == N:
        for line in board:
            print(" ".join("Q" if cell else "." for cell in line))
        return True

    for col in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1
            if solve(board, row + 1):
                return True
            board[row][col] = 0
    return False

board = [[0] * N for _ in range(N)]
solve(board, 0)
```

OUTPUT

One valid solution (Q = queen, . = empty square):

```
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```

RESULT

Thus, the 8 Queens problem was solved successfully using backtracking, placing all eight queens on the chessboard without any two queens attacking each other.